Validace OPL pravidly SHACL podle ISO 19450

OPL → RDF → SHACL validátor s hlášením chyb

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Ve zkratce

- Z ISO 19450:2024 jsem s Claude Opus 5.5 vytáhl **katalog 509 pravidel**. Každé z **533 „shall“** a 46 „should“ v textu normy v něm má záznam, i když se kontrolovat nedá.
- Co se z textu OPL kontrolovat dá, kontroluje můj nástroj `opmvalid`: **115 pravidel gramatika a 117 pravidel SHACL** (51 tvarů).
- Nástroj převede OPL na RDF, zkontroluje ho tvary SHACL a u každé chyby řekne **proč a kde**: kód, řádek, větu, článek ISO a radu, jak chybu opravit.
- Správnost dokládám druhým nezávislým výtahem, testy platných a neplatných vět, 18 modely ověřenými v OPCloudu a **shodou dvou nezávislých nástrojů SHACL** (pySHACL a Apache Jena).
- Hotový nástroj prověřil **nezávislý audit** silnějším modelem (Claude Fable 5.2): z 30 nálezů jsem 26 opravil.

1 Zadání a rozhodnutí

Zadání lze shrnout do pěti bodů:

1. z ISO 19450 vytáhnout všechna syntaktická a sémantická pravidla OPL, která jde,
2. převést je na pravidla SHACL,
3. napsat program, který převede OPL na RDF,
4. validovat RDF pomocí SHACL a hlásit, proč a kde není platné,
5. vyzkoušet záměrně platné a neplatné věty.

Výstupem má být kontrola, že pravidla SHACL opravdu odpovídají normě.

Na začátku jsem rozhodl několik věcí, ze kterých nástroj vychází:

- **Základ je norma.** Výchozí profil kontroluje podle ISO 19450. Volitelný profil OPCloud se přizpůsobí tomu, jak píše OPCloud (tabulka 3).
- **Závažnost podle normy.** „Shall“ je chyba (E), „should“ varování (W), jiné znění věty informace (I). Dvě pravidla jsem vědomě snížil na varování, protože nemění význam modelu (OPM-7-017 a OPM-14-091).
- **SHACL Core všude, kde stačí.** SHACL-SPARQL používám jen tam, kde Core nestačí, a důvod je u pravidla v katalogu.
- **Hlášení.** Jedna příčina dá jedno hlášení, anglicky, s článkem ISO a radou.

2 Pravidla z normy

2.1 Katalog

Jediným zdrojem pravidel je soubor `pravidla.yaml` (přes 10000 řádků). Každý záznam má kód (např. `OPM-9-036`), článek a stranu ISO, sílu (shall / should), kategorii a vlastní anglickou parafrázi pravidla. Normu necituji, je chráněná. U kontrolovatelných pravidel je v záznamu i tvar SHACL, text hlášení a rada. Tvary SHACL i texty hlášení se z katalogu generují.

Záznamy jsem rozdělil do pěti kategorií podle toho, čeho se pravidlo týká:

kategorie	záznamů	kontrolovatelné z textu OPL
1 syntaxe věty	120	ANO
2 struktura a význam modelu	125	ANO
3 grafika diagramu (OPD)	82	NE
4 chování za běhu	46	NE
5 definice a dovolení	136	NE
celkem	509	

Tabulka 1: Katalog podle kategorií.

Kategorie 1 a 2 kontroluje nástroj: 115 pravidel gramatika a 117 pravidel SHACL. Zbylých 13 pravidel z těchto kategorií nekontroluji a důvod je u každého v katalogu. Osm se v OPL nedá vůbec zapsat (např. rychlost spotřeby). U čtyř si norma sama odporuje nebo je nejasná. Jedno z modelu rozhodnout nejde.

2.2 Kontrola druhým, nezávislým výtahem

Výtah jsem po kapitolách nechal zopakovat samostatného agenta, který pracoval jen s textem normy a neviděl můj katalog. Oba výtahy našly **stejný počet výskytů „shall“ a „should“ ve všech článcích**.

Kde se výtahy lišily v kategorii nebo způsobu kontroly, jde většinou o styl: jestli definici vést jako samostatnou kontrolu, nebo ke kterému z několika „shall“ jednoho článku přiřadit společný tvar. Věcné rozdíly jsem prošel a zapracoval.

kapitola	6	7	8	9	10	11	12	13	14	A	B
shoda kategorie (%)	80	76	89	98	88	91	93	1/2	85	100	67
shoda kontroly (%)	91	82	84	97	90	91	93	1/2	84	100	40

Tabulka 2: Shoda katalogu s druhým výtahem.

2.3 Převod na gramatiku a SHACL

Syntaxi kontroluje gramatika podle Annex A: 115 pravidel, 114 z nich má test. Nečitelnou větu parser neopravuje. Nahlásí, kde čtení skončilo a jaký tvar věty to měl být.

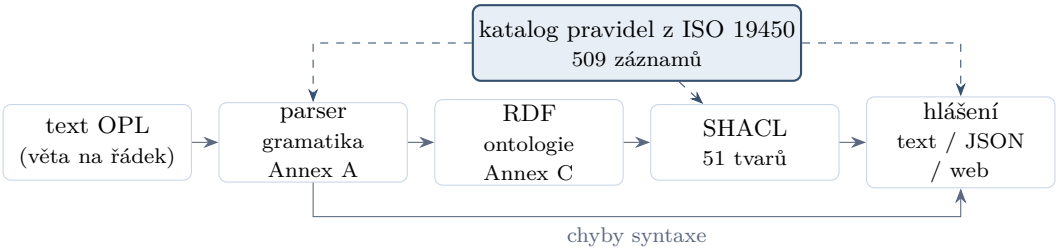
Strukturu a význam kontrolují tvary SHACL nad RDF modelem: 117 pravidel v 51 tvarech, **30 v SHACL Core a 21 v SHACL-SPARQL**. SPARQL je potřeba tam, kde se porovnávají hodnoty dvou různých uzlů (např. dvě vazby mezi týmž objektem a procesem), kde se počítá přes celý model nebo kde se z textu čte číslo (lhůta „5 minutes“).

Příklad tvaru v Core, pravidlo „agent je člověk“ (ISO 9.2.2, zkráceno):

```
opms:OPM-9-028 a sh:NodeShape ;
    sh:targetClass opm:AgentLink ;
    sh:property [ sh:path ( opm:object opm:isHuman ) ; sh:in ( true ) ] ;
    sh:severity sh:Violation .
```

** seznam všech tvarů je v příloze A.

3 Jak nástroj funguje



Obrázek 1: Cyklus OPL → RDF → SHACL → hlášení.

3.1 Převod OPL na RDF

Každá věc, stav, vazba i odstavec diagramu je v RDF samostatný uzel. Vazba si pamatuje větu a řádek, ze které vznikla, a proto hlášení umí říct, kde chyba je. Třídy a vlastnosti vycházejí z ISO 19450 Annex C.

Věta „Barista handles Brewing.“ dá mimo jiné tyto trojice (zápis Turtle):

```
m:l1 a opm:AgentLink ;
    opm:object m:t_Barista ; opm:process m:t_Brewing ; opm:sentence m:s1_1 .
m:s1_1 opm:line 1 ; opm:text "Barista handles Brewing" .
m:t_Barista a opm:Object ; opm:name "Barista" ; opm:isHuman true .
```

Údaj `opm:isHuman` v textu OPL není. Norma přitom chce, aby agent byl člověk, proto ho dodává volitelný seznam lidí. Bez seznamu se pravidla o agentovi a nástroji nekontrolují a nástroj to oznámí. ** adresy `example.org` jsou zatím zástupné.

3.2 Hlášení

Pro ukázkový model se schválnými chybami (12 řádků) nástroj vypíše 9 chyb, 1 varování a 2 informace. Dvě z chyb:

```
E OPM-9-036   line 7, col 1: State not possible: "Printing yields final Report" uses the
              state "final", which is not a declared state of "Report".
    [ISO 9.3.1]  Fix: Declare it ("Report can be ... or final.") or use one of the declared
                 states of "Report": approved, draft.
E OPM-9-028   line 11, col 1: "Printer" handles "Printing", but "Printer" is not a human: an
              agent must be a human or a group of humans.
    [ISO 9.2.2]  Fix: If "Printer" is a system or a device, write "Printing requires Printer."
                 (instrument link).
```

3.3 Použití

Nástroj se spouští z příkazové řádky:

```
python -m opmvalid model.opl --profile iso --format text --humans "Operator" --turtle model.ttl
```

- `--profile`: `iso` (výchozí), nebo `opcloud`,
- `--format`: `text` (výchozí), nebo `json`,
- `--humans`: seznam lidí v modelu, oddělený čárkou (nepovinný),
- `--turtle`: uloží model v RDF do souboru (nepovinný).

Návratový kód: 0 bez chyb, 1 chyby, 2 vstup nejde přečíst.



Obrázek 2: Webové rozhraní: ukázkový model se schválnými chybami, výsledek validace a začátek modelu v RDF.

3.4 Profily ISO a OPCloud

OPCloud píše některé věty jinak než norma a má dvě vlastní pravidla. Profil OPCloud je proto norma tak, jak ji používá OPCloud. Gramatika a 48 z 51 tvarů SHACL jsou v obou profilech stejné. Liší se jen tohle:

kontrola	profil ISO	profil OPCloud
věta ve znění OPCloudu (lhůty, výjimky, vzájemný vztah)	I	-
vypsaná výchozí vlastnost věci (OPM-7-017)	W	-
chybějící titulek specifikace (OPM-14-078)	W	-
vazby téhož druhu v oddělených větách (OPM-12-002, N-062)	I	E
víc koncových stavů objektu (OPC-001)	-	E
physical podproces v informatical procesu (OPC-002)	-	E

Tabulka 3: Rozdíl profilů. E chyba, W varování, I informace, - nekontroluje se.

4 Doklad, že pravidla odpovídají normě

zkouška	výsledek
pokrytí normy (výskyty „shall“ / „should“ se záznamem)	533/533, 46/46
druhý nezávislý výtah (stejné počty výskytů po článcích)	všechny články
platné věty	169/169
neplatné modely	132/132
nová doména (model psaný až po testech)	29/29
věty modelů z OPCloudu, které nástroj přečte	1399/1399
dva nástroje SHACL, pySHACL a Apache Jena (shodné výsledky)	168/168 modelů
web a API proti příkazové řádce	46/46
nezávislý audit (Claude Fable 5.2)	30 nálezů, 26 opraveno

Tabulka 4: Zkoušky sestavené s pomocí Opus 5.5, audit Fable 5.2.

Co jednotlivé zkoušky dokládají:

- **Pokrytí normy.** Skript spočítá slova „shall“ a „should“ v textu normy po článcích a porovná je se záznamy katalogu. Žádné nechybí.
- **Platné věty.** 169 vět ve všech tvarech normy s vlastními názvy. Každá se musí přečíst jako správný druh věty.
- **Neplatné modely.** 132 modelů: 99 porušuje pravidla, 33 je schválně platných. Nástroj musí nahlásit přesně očekávané kódy a nic navíc, tedy bez falešných nálezů a kaskád. Každý tvar SHACL má svůj případ (47) nebo zdůvodnění, proč ho mít nemůže (4).
- **Nová doména.** Model o půjčování knih jsem psal až po ostatních testech a očekávané kódy zapsal předem. Napoprvé prošlo jen 23 z 28 bodů a zkouška našla tři vady, které předchozí testy neviděly. Všechny jsou opravené a mají test (dnes má zkouška 29 bodů).
- **Modely z OPCloudu.** 18 referenčních modelů ověřených v OPCloudu. Nástroj přečte každou větu a všechny jeho nálezy jsem ručně prošel (oddíl 5).
- **Dva nástroje SHACL.** Stejné tvary a stejná data jsem pustil v pySHACL a v Apache Jena. Na 168 modelech dávají stejných 821 výsledků, tvary jsou tedy standardní SHACL.

- **Nezávislý audit.** Hotový nástroj prověřil samostatně silnější model (Claude Fable 5.2), který neviděl průběh práce. Psal vlastní modely naslepo s očekáváním zapsaným předem, prošel všech 51 tvarů proti normě a vzorek 40 záznamů katalogu. Našel 30 nálezů, hlavně méně časté tvary vět z příkladů normy, podmínkové věty a vztahy mezi diagramy. 26 jsem opravil a každá oprava má test. Hlášení na modelech z OPCloudu se opravami nezměnila a druhé kolo naslepo na opravené verzi vyšlo ve 41 ze 42 modelů podle očekávání.

Ke každému pravidlu lze jedním příkazem vypsat celý řetěz od normy po test:

```
python katalog/pravidlo.py 9-036 --run
```

Vypíše parafrázi, věty normy, tvar SHACL a testy s výsledkem. Slouží k namátkové kontrole.

5 Nálezy a omezení

5.1 Modely z OPCloudu

Na 18 referenčních modelech dávají oba profily stejných **7 chyb** a všechny jsou oprávněné:

- 5× OPM-14-054: událost zvenku spouští podproces rozbaleného procesu mimo pořadí,
- 1× OPM-14-068: v SD jsou dva systémové procesy,
- 1× OPM-14-086: vazba v SD neodpovídá nejsilnější vazbě k podprocesům.

Varování: 6× diagram s víc než 25 věcmi (OPM-B-002), 2× objekt vytvořený dvakrát (OPM-14-088) a 1× modifikátor při abstrakci (OPM-14-091).

Profil ISO navíc hlásí 695× vypsanou výchozí vlastnost, 18× chybějící titulek a 144 vět ve znění OPCloudu. OPCloud se tedy v zápisu od normy odchyluje, význam modelů je ale v pořádku.

5.2 Nálezy v normě

Při výtahu jsem v ISO 19450:2024 zapsal 62 nálezů (6 vysoké, 32 střední a 24 nízké váhy, příloha B). Typicky jde o rozpor mezi textem článku, gramatikou v Annex A a příklady:

- Annex A píše větu agenta se slovesem „handle“, text a příklady „handles“ (N-015).
- Výčet stavů spojuje Annex A s „and“, příklady s „or“ (N-010).
- Podmínku agenta a podmínku nástroje Annex A z textu nerozliší (N-017).
- Článek 12.1 žádá vazby téhož druhu v jedné větě, tři obrázky normy je píšou odděleně (N-062).

Kde norma uvádí dvě verze, profil ISO přijímá obě. U věty, která odpovídá jen jedné z nich, dá informaci.

5.3 Omezení

- **Grafika a chování za běhu.** Pravidla o vzhledu diagramu (82) a o simulaci (46) se z textu OPL kontrolovat nedají.
- **Co OPL nevyjádří.** 13 pravidel struktury a syntaxe nekontroluji (oddíl 2.1). Pravidla o agentovi a nástroji potřebují seznam lidí, který v textu OPL není.
- **Jednotné číslo názvu (OPM-B-005).** Toto doporučení není hotové. Bez knihovny na tvary anglických slov by dávalo plané nálezy (Process, Glass, Analysis).

A Tvary SHACL

Všech 51 tvarů ze souboru `opmvalid/tvary.ttl` (30 v SHACL Core, 21 v SHACL-SPARQL). Další pravidla katalogu, která se kontrolují tvarem SHACL, sdílejí některý z těchto tvarů. Parafráze pravidel jsou z katalogu (anglicky, jako hlášení nástroje).

kód	ISO	SHACL	záv.	profil	pravidlo (parafráze)
OPM-9-001	9.1.1	Core	E	oba	A transforming link connects a process with an object, its transformee: one end is a process, the other end is an object.
OPM-9-021	9.2.1	Core	E	oba	An enabling link connects an enabler, which is an object, with the process it enables.
OPM-9-028	9.2.2	Core	E	oba	An agent is a human or a group of humans able to make intelligent decisions, who enables or controls the process.
OPM-9-032	9.2.3	Core	E	oba	An instrument is an inanimate or otherwise non-decision-making enabler; the process cannot start or run without it.
OPM-9-036	9.3.1	SPARQL	E	oba	A state-specified consumption link goes from a specified state of the consumee to the process; the state is a state of that object.
OPM-9-051	9.3.2	Core	W	oba	A result link to an object with an initial state should end at the object itself or at a state other than the initial one, not at the initial state.
OPM-9-054	9.3.3.2	Core	E	oba	An input-output-specified effect link is a pair: from an input state of the affectee to the process, and from the process to a different output state of the same affectee.
OPM-9-087	9.5.1	Core	E	oba	An exception link connects a process that cannot complete normally (or in the expected time) with a process that handles the exception.
OPM-9-170	9.5.4.1	Core	W	oba	The Minimal Duration of a process does not exceed its Maximal Duration (they are the minimum and the maximum allowed time for completing the process).
OPM-10-001	10.1	Core	E	oba	A structural link connects objects with objects or processes with processes, never an object with a process, except the exhibition-characterization link.
OPM-10-020	10.3.1	Core	E	oba	A fundamental relation refines exactly one source thing (refineable) into one or more destination things (refinees).
OPM-10-037	10.3.3.1	Core	E	oba	An attribute is a feature that is an object.
OPM-10-050	10.3.3.2.2	SPARQL	E	oba	A sentence about a feature or its value identifies the feature's exhibitor („Feature of Exhibitor“); the exhibitor must actually have that feature, directly or by inheritance.
OPM-10-068	10.3.4.3	SPARQL	W	oba	With several discriminating attributes, the number of specializations is at most the product of the numbers of their values.
OPM-10-070	10.3.5.1	SPARQL	E	oba	If the class gives an attribute a range of values, the value of that attribute for each instance lies within the range.
OPM-10-085	10.4.1	Core	E	oba	The specialization then has only that value of the inherited attribute.
OPM-11-023	11.1	Core	W	oba	In a multiplicity range qmin..qmax the lower bound does not exceed the upper bound.
OPM-12-002	12.1	SPARQL	E	oba	Links with AND semantics are written as one OPL sentence joining the things with „and“, not as one sentence per link.
OPM-12-022	12.7	Core	E	oba	In a probabilistic link fan every link of the fan carries a probability.
OPM-12-023	12.7	SPARQL	E	oba	The probabilities of a probabilistic link fan add up to exactly 1.
OPM-7-006	7.2.1	SPARQL	E	oba	A process has a positive duration.
OPM-7-009	7.3.1	Core	E	oba	A thing is either an object or a process.

kód	ISO	SHACL	záv.	profil	pravidlo (parafráze)
OPM-7-017	7.3.4	SPARQL	W	ISO	For a thing linked to other things, the OPL does not state default values of its generic properties.
OPM-7-023	7.3.5.3	Core	E	oba	An object has any number of initial and final states but at most one default state.
OPM-8-002	8.1.2	Core	E	oba	Every process is connected by a transforming link to at least one object or object state.
OPM-8-004	8.1.2	SPARQL	E	oba	At one extent of abstraction an object or object state links to a process by only one procedural link.
OPM-8-008	8.2.2	Core	E	oba	Every process has a preprocess object set with at least one object: something it consumes, affects or needs as an enabler.
OPM-8-010	8.2.2	Core	E	oba	Every process has a postprocess object set with at least one object: something it yields or affects, or an enabler (enablers and affectees belong to both sets, NOTE 1).
OPC-001	-	Core	E	OPCloud	An object has at most one final state (OPCloud; ISO allows several, 7.3.5.3).
OPC-002	-	Core	E	OPCloud	A physical subprocess cannot be inside an informatical in-zoomed process (OPCloud).
OPM-13-001	13	SPARQL	E	oba	A path label is a link property that pairs procedural links: the label of a link into a process is matched by the same label on a link out of it.
OPM-14-020	14.2.1.2	SPARQL	E	oba	The result of unfolding is a hierarchy tree: a refinement hierarchy has no cycles.
OPM-14-050	14.2.2.4.1	SPARQL	W	oba	An enabler attached to the contour of an in-zoomed process connects to at least one of its subprocesses.
OPM-14-051	14.2.2.4.1	SPARQL	E	oba	Consumption and result links are not attached to an in-zoomed process; they go to its subprocesses.
OPM-14-054	14.2.2.4.2	SPARQL	E	oba	An event link does not cross the boundary of an in-zoomed process from outside to initiate one of its subprocesses (at any level).
OPM-14-055	14.2.2.4.3	SPARQL	E	oba	After in-zooming a process that changes an object from one state to another, the input and output links go to subprocesses in a temporally feasible way (one subprocess, or the input to an earlier one and the output to a later one).
OPM-14-061	14.2.2.6.1.1	Core	E	oba	The OPD process tree is a directed tree with the SD as root and the other OPDs as nodes.
OPM-14-062	14.2.2.6.1.1	Core	E	oba	Each edge of the OPD tree goes from the parent OPD holding the refineable to the child OPD with its refinees: every OPD except the SD refines exactly one thing of exactly one parent OPD.
OPM-14-094	14.2.2.6.1.1	SPARQL	E	oba	The parent OPD of an edge of the OPD tree contains the refineable: the thing that a child OPD refines appears in the parent OPD that the refinement sentence names.
OPM-14-067	14.2.2.6.1.3	Core	E	oba	Tier 0 has exactly one OPD, the SD.
OPM-14-068	14.2.2.6.1.3	Core	E	oba	The SD contains one and only one systemic process, the overarching system function; environmental processes may be added.
OPM-14-069	14.2.2.6.1.4	SPARQL	E	oba	Each edge label of the OPD tree refers to the refinement of a uniquely named process into another OPD.
OPM-14-081	14.2.3	Core	E	oba	No OPD contains a fact that contradicts a fact of the same or another OPD (e.g. „P yields A.“ and „P consumes A.“); a refinement of a fact is not a contradiction.
OPM-14-086	14.2.4.1	SPARQL	E	oba	The relative semantic strength decides which link prevails when an OPD is abstracted: the link between an object and a refined process is the strongest of its links to the subprocesses.

kód	ISO	SHACL	záv.	profil	pravidlo (parafráze)
OPM-14-088	14.2.4.1.1	SPARQL	W	oba	Table 27: within an in-zoomed process an object cannot be consumed and then consumed or affected by a later subprocess, nor affected or created and then created by a later one.
OPM-14-093	14.2.2.6.1.4	Core	W	oba	Each elaborated process has one refinement into another OPD: a process is in-zoomed into one set of subprocesses, not two different ones.
OPM-14-091	14.2.4.1.3	SPARQL	W	oba	Within each kind, an event link is stronger and a condition link weaker than the basic link.
OPM-B-002	B	Core	W	oba	An OPD should contain no more than 20-25 things (B.3).
OPM-B-006	B	Core	W	oba	A tool should notify the modeller when an object is a refinee in more than one context (B.6.2).
OPM-B-007	B	Core	W	oba	A process name should end with a gerund (-ing) and have no more than four capitalised words (B.6.3).
OPM-B-008	B	SPARQL	W	oba	State names should describe situations, preferably as passive forms (painted), not gerunds (painting) (B.6.4).

B Nálezy v normě

Při výtahu pravidel jsem v ISO 19450:2024 zapsal 62 nálezů: rozpory mezi textem článku, Annex A a příklady, chyby v obrázcích a příkladech, nejasnosti. Každý je v katalogu s místem v normě; popisy jsou z katalogu (anglicky, parafráze, bez delších citací normy).

	místo v ISO	váha	popis
N-001	9.1.1, Figure 7	střední	The captions of Figure 7 do not match the diagrams and sentences: a) is labelled Result but shows a consumption („Deleting consumes File“), b) is labelled Effect but shows a result („Creating yields File“), c) is labelled Consumption but shows an effect („Editing affects File“).
N-002	9.1.3, EXAMPLE 2	nízká	The example computes the output of Refining and ends with „8,000 cubic meters of Crude Oil“; by the rest of the example it should be Diesel Oil (Crude Oil is the consumee).
N-003	9.1.1, Figure 7	nízká	The OPL sentences under Figure 7 have no final full stop, which Annex A requires for every OPL sentence; Table 1 (9.1.5) writes them with it.
N-004	Annex A.4.3, „object with optional state“	střední	The production puts a mandatory space between the optional state and the object identifier. Without a state, a literal reading gives a double space (e.g. „Eating consumes Food“). The grammar implementation has to treat the space as part of the optional state.
N-005	Annex A.4.5.2.2-4, Xor sentences with several processes	střední	„Exactly “ followed by „process Xor list at beginning“, which starts with „One of “, yields „Exactly One of ...“ with a capital O in the middle of the sentence (consumption destination Xor, result source Xor, effect process Xor, agent source Xor, instrument destination Xor).
N-006	Annex A.4.3, „object nostates Xor list at end“	střední	The production is named „nostates“ but is defined with „object Or list“ (objects with optional states) instead of „object Or list nostates“. Used by the effect Xor sentence, where states are not allowed.
N-007	Annex A.4.5.2.4	nízká	The effect object Or sentence refers to „object Or list Nostates“ (capital N), which is not defined (the list is „object Or list nostates“). The cross-reference comment of the effect sentence says „see 9.14“ instead of 9.1.4.
N-008	9.1.2, 9.1.3 versus Annex A and clause 11	střední	The body requires a rate property on consumption and result links that model transformation over time, but Annex A defines no OPL sentence for link properties such as a rate. These requirements are therefore not visible in OPL text.
N-009	9.1.4 versus Table 1	nízká	9.1.4 defines the effect as an unspecified change; Table 1 describes it as a change from one state to another. Read together with 9.1.4, the catalogue does not require states for an affectee.

	místo v ISO	váha	popis
N-010	9.3.2, Figure 10, versus Annex A.4.4.4	vysoká	Annex A joins the last state of a state enumeration with „and“ („A can be s1, s2 and s3“), but every example in the body joins it with „or“ („A can be s1, s2, or s3.“, „Raw Metal Bar can be pre-cut or cut.“, „Moving Truck can be worn out or serviced.“), as OPCloud does; chapter 10 mixes both („Gender of Adult can be female or male.“ in Figure 26, „... can be ground, air, and water surface.“ in Figures 25 and 28). Figure 10 also writes „S2 is initial.“, a form Annex A does not define (it has „State s2 of A is initial.“). The state declaration is one of the most frequent OPL sentences, so the ISO profile has to decide which form it accepts.
N-011	9.3.3.2, Figure 11	střední	The diagram of Figure 11 draws the consumption of Raw Metal Bar from its state cut, and the text says Machining consumes it only in that state, but the OPL under the figure says „Machining consumes Raw Metal Bar.“ without the state. The environmental process Cutting (dashed) has no OPL sentence declaring it environmental, while Testing has one.
N-012	Figures 11 and 12 versus Annex A.4.4.2	střední	The examples write generic properties as „Raw Metal Bar is physical.“ and „Testing is environmental and physical.“ (Figures 11, 12), Figure 4 as „Physical Systemic Object is a physical and systemic object.“. Annex A defines the values with capitals („Physical“, „Environmental“), in the order essence, affiliation, perseverance, and with no separator between them, so a literal reading gives „Testing is PhysicalEnvironmental“.
N-013	9.3.3.4	nízká	In the clause on the output-specified effect link, the syntax sentence is introduced as „the syntax of an input-specified effect link“; the form given („changes Object to output-state“) is the output-specified one.
N-014	Annex A.4.2 versus 9.4.2, Tables 3, 7, 13	střední	Annex A defines a state identifier as a single non-capitalised word, but the body uses multi-word states: „worn out“, „old apartment“, „awaiting test“, „passed test“, „safety design authorized“, „periodically calibrated“. The examples also capitalise a state that starts a sentence („Sober Pilot handles Flying.“, „Awaiting test Sample initiates Testing, ...“), as the NOTE in 9.4.1 prescribes, which a non-capitalised identifier cannot produce. Attribute values such as 75 (7.3.5.5) are states too, but a number is hardly a „non capitalized word“. A parser that follows Annex A literally rejects the norm's own examples.
N-015	Annex A.4.5.3.2, agent sentence	vysoká	The basic agent sentence of Annex A uses the verb „handle“ (object with optional state list, „ handle “, process identifier), which gives „Welder handle Welding“. The body (9.2.2) and all examples write „handles“. The plural form would fit a list of agents, but the production does not distinguish one agent from several.
N-016	Annex A.4.5.3.2, agent Or sentences	střední	The agent source Or sentence has „handles“ without spaces („... object Or list, “handles„, process ...“), giving „...or Bobhandles Welding“; the agent destination Or sentence has „handles at least one of “ without the leading space.
N-017	Annex A.4.5.4.3 versus 9.5.3.2.1	vysoká	In Annex A the conditional agent sentence and the conditional instrument sentence are the same production („<Process> occurs if <Object> exists, else <Process> is skipped“), so from OPL text an agent condition cannot be told from an instrument condition. The body gives the agent its own form („<Agent> handles <Process> if <Agent> exists, else <Process> is skipped“).
N-018	9.5.3, Tables 10, 11, 13, Figure 13, Annex A.4.5.4.3	vysoká	The wording of condition sentences differs between the body, the tables and Annex A. Consumption: body „in which case Object is consumed“, Table 10 „in which case Process consumes Object“. Agent and instrument: body „else“, Tables 11 and 13 and Figure 13 „otherwise“; Table 11 „if Engineer is present“ instead of „exists“. State-specified effects: body „occurs if Object is input-state“, Annex A „occurs if there is input-state Object“ and „else“ instead of „otherwise“. The second forms of the state-specified effect and instrument sentences („If input-state Object then ...“) lack the „exists“ that the consumption form has, and the effect forms also drop „occurs and“. The Annex A alternatives of the output-specified effect and of the state-specified instrument condition differ from the body forms as well („... in which case P changes O to s, otherwise bypass P“; „P occurs if s I exists, else ...“).

	místo v ISO	váha	popis
N-019	Annex A.4.5.4 (event, condition, invocation sentences)	střední	Literal reading of the control productions gives malformed strings: „occurs if “ and „occurs and affects “ without the leading space (simple conditional effect sentence); „is skipped “ with a space before the full stop; „input state, object identifier“ without a space between state and object (effect event and conditional sentences); „invokes at least one of“ without the trailing space; „invokes itself “ with a trailing space; no concatenation comma before „object with optional state“ in the instrument event sentence; „Input state“ capitalised (undefined); no comma before „in which case“ in the input specified conditional effect sentence. The invocation Xor sentence writes „Exactly one of“ in lower case, unlike the other Xor sentences (N-005). The conditional effect sentence lists the alternative „output specified condition effect sentence“, which is not defined (the production is „output specified conditional effect sentence“). In the conditional consumption sentences the literal „is consumed, otherwise“ is broken across two lines inside the quotes, so a literal transcription contains a line break and indentation.
N-020	Annex A.4.5.2.5, change sentences	nízká	Production names do not match their uses: „in out specified change Object Xor sentence“ is defined with a capital O but referred to as „... object Xor sentence“; „Or In out object change list“ versus „Or in out object change list“. The output object change list misses the concatenation comma („, “ output object change phrase). In the output specified change Or sentence the alternative with several processes takes a whole output object change list, where the in-out and input analogues take a single change phrase.
N-021	9.5.2.3.4, Table 7	střední	The output-specified effect event link starts from the object in any state („Object in any state initiates ...“), but the example in Table 7 draws the event from the state rusty and writes „Rusty Engine Hood initiates Cleaning & Painting which changes Engine Hood to painted“ (a specified state, no comma, no full stop).
N-022	9.5.3.4.3, Table 13	nízká	One example names the same things twice differently: the diagram has „LASER Metre“, the OPL „LASER Meter“; the OPL calls the process „Ultra-Precision Measuring“ and, in the same sentence, „Ultra-Precise Measuring“.
N-023	9.3.2, NOTE 2 versus Figure 10	nízká	The rule on result links to an initial state is only a recommendation („should“) and NOTE 2 says the modeller can want the right-hand OPL, yet Figure 10 labels that variant „Incorrect“; 7.3.5.3 defines the initial state as the state upon generation, which makes a result link to it natural. The catalogue keeps the weaker reading (warning W, OPM-9-051).
N-024	9.5.3.3.5, Table 12	střední	For the condition output-specified effect link, the semantics column of Table 12 says the process changes the object from its input state to any one of its states, while 9.5.3.3.4 and the sample sentence of the same row say it changes the object to the specified output state.
N-025	Annex A.4.5.2.5, change Or / Xor sentences with several processes	střední	The change select sentences with several processes have no „At least one of “ (Or) and no „Exactly “ (Xor) prefix („process Or list, “ changes „, ...“; „process Xor list at beginning, “ changes „, ...“), unlike the consumption, result, effect and instrument select sentences, so the Or and Xor variants read alike apart from „One of“.
N-026	9.3.3.3	nízká	„Existence of the affectee at the input source state is a precondition ...“ is stated without „shall“, unlike the parallel sentences of 9.3.3.2 and 9.3.3.4. Requirements can therefore be phrased without the modal verb; the catalogue records this one as OPM-9-169 (strength implicit), outside the shall count.
N-027	chapter 9, editorial	nízká	Editorial slips that do not change a rule: „preforming the Sintering process“ for performing (9.2.3, EXAMPLE 2); „default state. then“ with a stray full stop (9.3.3.3); Figure 11 captioned „results links“ and placed in the effect-link clause; „initiates process“ for initiated process in Table 7; a stray closing quotation mark after the syntax of 9.5.2.4.2; „shall be: either“ with a misplaced colon (9.5.3.3.4); circular graphical definitions in 9.5.3.3.2-9.5.3.3.4 („the condition ... link with the letter c shall denote the condition ... link“).

	místo v ISO	váha	popis
N-028	10.3.4.1, Figure 24, Annex A.4.6.5	střední	10.3.4.1 defines generalization as a general with two or more specializations, while the next sentence of the same clause speaks of one or more, 10.3.1 of one or more refinees, Figure 24 shows single specializations („Digital Camera is a Camera.“, „Hunting is Food Gathering.“) and Annex A has a sentence for one. The catalogue does not report a single specialization.
N-029	10.3.3.2.2 EXAMPLE 1, 10.3.5.1 EXAMPLE 2, Figure 27	střední	The OPL and the text give the range of Specific Weight as „from 7,545 to 7,537“ (lower bound above the upper one) and the instance value as 7,555, which the text calls „within the allowable range“; the diagram shows 7,545..7,573 and 7,55. The example meant to illustrate the range rule of 10.3.5.1 (OPM-10-070) breaks it in its OPL. The text calls the instance „Metal Powder Mixture Instance“, the figure „Mixture Lot #7545“.
N-030	10.3.5.1, Figure 26	nízká	In one figure: the diagram gives Height of Jack Robinson as 186, the OPL and the text as 185; the OPL writes „Weight in Kg of Adult range from 40 to 240.“ („range“ for „ranges“, „Kg“ next to „kg“).
N-031	10.3.3.1 versus Annex A.4.6.3.2	střední	Operations of an exhibitor are called „Operation“ and „Operator“ within one syntax template („Operation1, Operator2, ..., Operatorm“); the incomplete forms say „and at least another operation“ but „and at least one other attribute“; Annex A uses „operator list“ and „at least one other operator“. The partial characterization literals end with a space („, and at least one other attribute “), and in the AsWellAs process sentence the literal „, as well as “ is broken across lines.
N-032	chapter 10 examples versus Annex A.4.3 lists	střední	The examples write lists with the serial comma („Subject, Predicate, and Object“, „Car, Aircraft, and Ship are Vehicles.“, „Gender, Height in cm, and Weight in Kg“), while every list of Annex A joins the last item with „ and “ and no comma. The grammar accepts both.
N-033	10.3.3.2.2, 10.3.4.3, 10.4.1 versus Annex A.4.6.3-A.4.6.4	vysoká	Annex A has no production for the attribute value enumerations the examples use („Travelling Medium of Vehicle can be ground, air, and water surface.“, „Gender of Adult can be female or male.“): the state enumeration sentence takes a plain object identifier, without „of Exhibitor“. It also has no production for the state-specified characterization sentence of 10.4.1 („Car exhibits ground Travelling Medium.“), since its attribute list takes objects without states. The names are reversed as well: Annex A calls the „exhibits“ sentences characterization sentences and the „Feature of Exhibitor“ sentences exhibition sentences.
N-034	Annex A.4.6 (structural sentences)	střední	Literal reading gives malformed strings: biDirNullTag „ related“ after „ are “ (double space); „ and “, „ at least one other part“ (double space); general process list starts with „ a“ without a space after it and misses the comma before „ and a “; the multiple process instantiation sentence writes „are an instance of“ where the object one writes „are instances of“; the partial specialization sentence appends „ and other specializations“ to a list that already ends with „ and “ („A, B and C and other specializations are G“). Annex A also has specialization forms with no syntax in chapter 10 („can be either ... or ...“, „is a G1 and a G2“, state specialization); they are accepted.
N-044	10.2.4	nízká	The default reciprocal tag is given as „are related“, while the reciprocal syntax already supplies „are“ („Source and Destination are reciprocity-tag“), so a literal reading gives „are are related“; Table 14 and Annex A use the tag „related“.
N-045	10.4.2.9, Figures 29 and 30, versus 10.3.3.2.2	nízká	The examples enumerate attribute values without the exhibitor („Keeper can be payer, payee, or financial institution.“, „Phase can be solid, liquid, or gas.“), although 10.3.3.2.2 requires „of Exhibitor“ when values of an attribute are expressed (Figure 25 writes „Travelling Medium of Vehicle can be ...“).
N-046	14.2.1.1, Figure 45, versus Annex A.4.4.4	střední	Suppressed states are written „or other states“ („A can be s1, s3, or other states.“), Annex A writes „ and other states“.

	místo v ISO	váha	popis
N-047	14.2.2.6.1.4, Table 26, versus Annex A.4.7	střední	The body expresses new-diagram refinement as „SD is refined by in-zooming Dish Washing in SD1.“ (an OPD refinement sentence); Annex A has no such sentence and writes instead „<Process> from SD zooms in SD1 into ...“, which is what OPCloud writes. The syntax templates of 14.2.2.6.1.4 also have misplaced quotation marks. Annex A has no production for the document frame of Table 26 („OPL specification of ...“, paragraph labels „SD1: ...“, „End of OPL specification of ...“).
N-048	14.2.1.3, 14.2.2.1-14.2.2.3, Figures 46, 47, 49, Table 24, Annex A.4.7.4	střední	The in-zoom sentence is written with „, in that sequence“ (clauses, Table 24, Figure 48) and without the comma (Figures 46, 47); Annex A concatenates process list and „in that sequence“ without comma or space („... and Cin that sequence“). Parallel subprocesses are written „zooms into parallel A and B“ (14.2.2.2), „A, parallel B and C, D, and parallel E, F, and G, in that sequence“ (Figure 49), „A and parallel B and C, in that sequence“ (Table 24); Annex A allows „parallel“ only at the start or before a last group. Annex A names „process out Zoom sentence“ with a capital Z but refers to „process out zoom sentence“.
N-049	13, 14.2.1.2, 14.2.2.5, 14.2.4.1.1, Figures 47, 53, 54, Table 26	nízká	Editorial and naming slips in examples: „Dish Washer consists of Soap Compartment“ next to „Dishwasher“ (Figure 53, Table 26); „Detection Treat“ next to „Detected Threat“ and a sentence without its full stop (Figure 54); „Refine2“ in the unfolding template (14.2.1.2); the text of 14.2.4.1.1 names „B-to-P1 Link“ for both the column and the row (the row is B-to-P2); „Instrument“ capitalised in the order list (14.2.4.1.4). In Figure 47 nothing takes Keeper from payee to financial institution, although Cashing & Cancelling changes it from financial institution to payer. 14.2.2.4.2 ends with a paragraph about „the skipped process“ that nothing in the clause introduces.
N-050	14.2.4.1.2 versus 14.2.2.4.3 NOTE 2 and Figure 53	střední	By the precedence rules transforming links prevail over enabling links when an OPD is abstracted, but NOTE 2 and Figure 53 keep Dishwasher an instrument of Dish Washing in the SD although the subprocesses in SD1 change it (empty to loaded and back to empty). The catalogue applies the precedence with that exception (OPM-14-086).
N-058	14.2.4.1.1, Table 27, versus 14.2.2.4.4	nízká	Table 27 gives an effect link to the out-zoomed process when an earlier subprocess yields B and a later one consumes it; then B exists neither before nor after the process (an interim object, 14.2.2.4.4), so it is not affected from outside.
N-059	14.2.2.4.3 versus Figure 52, Table 25	nízká	The clause says to attach both the input and the output link „to one of the subprocesses“, while the remedy it then defines, Figure 52 c) and Table 25 split them between two subprocesses. The catalogue accepts both (OPM-14-055).
N-051	Figures 47, 53, Table 26, versus Annex A.4.4.4	střední	The examples write „State payer of Keeper is initial and final.“ and „State empty of Dishwasher is initial and final.“; Annex A has separate initial and final sentences and a combined „is initially ... and finally ...“ form, but no „is initial and final“. OPCloud writes the same combined form as the examples.
N-052	7.3.3, 7.3.4, 10.3.1 versus Annex A.4.4.2	střední	Perseverance has the values „static, dynamic or persistent“ in 7.3.3 (objects static, processes dynamic; the NOTE adds persistent and transient), „static / dynamic“ in 10.3.1, and „Persistent | Transient“ with default Persistent in Annex A. The default Essence is the system's Primary Essence (majority of things) in 7.3.4, but „Informatical“ in the comment of Annex A.
N-053	7.3.4 versus Figures 11, 12	nízká	7.3.4 says the generic properties of a linked thing merge into the OPL phrases of its links; the examples keep separate sentences for linked things („Raw Metal Bar is physical.“, „Moving Truck is physical.“).
N-054	11.2, Figure 33, versus 8.2.2	nízká	„Purchasing yields many new Blades.“ gives a process with no preprocess object (nothing consumed, affected or enabling), which 8.2.2 requires for every process; so do the one-sentence examples „Creating yields File.“ (Figure 7) and „Mining yields Copper.“ (Table 1), and „Deleting consumes File.“ has no postprocess object. Fragments, but 8.2.2 is stricter than 8.1.2, which asks only for one transforming link.
N-055	6.2.6.3 versus 14.2.2.6.1.3-14.2.2.6.2	střední	6.2.6.3 says that clause 14 does not specify the labels of hierarchic levels, of links between OPDs or of OPL segments; clause 14 does specify the SD label, the edge labels and refinement sentences of the OPD tree and the OPL paragraph labels (Table 26).

	místo v ISO	váha	popis
N-060	8.1.2 versus 9.3.3, 9.5.2	střední	8.1.2 gives an object exactly one role per process (transformee, enabler, initiator or conditional object) and allows only one procedural link between them. An agent event link makes the agent both initiator and enabler („Agent initiates and handles Process.“), and an input-output-specified effect link is defined as a pair of links (9.3.3). The catalogue reads a control modifier and the two halves of a change as parts of one link (OPM-8-004).
N-035	10.4.2.7	nízká	The reciprocal source-or-destination state-specified tagged link without a tag has a default tag („are related“) but no syntax sentence, unlike 10.4.2.8, which gives both.
N-036	chapter 10, editorial	nízká	„Folding the refines“ for refinees (10.3.1); syntax sentences ending in a semicolon instead of a colon („shall be;“, 10.3.5.1); „in in Figure 28“ (10.4.1); „relations and inks“ (10.4.2.9); a graphical sentence without its full stop (10.3.4.1). Figure 29: EXAMPLE 1 lists the values of Keeper as payer, payee and bank, while the OPL and the end of the same example say financial institution.
N-037	11.1-11.3 versus Table 16, Figures 32-34, 26, Annex A.3.2	střední	A range „..“ shall read „through“ in OPL, but the examples and Annex A write „to“ („3 to 5 or 8 to 10 Machines“, „k=2 to 4“; OPCloud too). The range sentence shall be „Attribute of Object range is value-range“, the examples and Annex A write „... ranges from <min> to <max>“. The plural rule is broken by the examples: „Car carries at least one Spare Tire.“ (several possible, singular) and „b Engines“ with b in {0, 1}. Figure 33 writes constraints inline („k=2 to 4“, „a <= b inspected Blades“) instead of „, where ...“, uses „many“, which clause 11 does not define, and names a parameter „a“, like the article. The semicolon-separated constraint list of 11.2 concerns the OPD annotation, so Figure 34's commas in OPL are no contradiction.
N-038	Annex A.1, A.3.2 versus 11.1-11.3	střední	Annex A states itself that it is incomplete: no productions for probabilities (12.7), execution paths (clause 13) and most participation constraints. It defines only a single range „<limit> to <limit>“, the phrases of upper single / lower plural („an optional“, „at least one“, „optional“, „many“) and a constraint „ where <name> <operator> <value>“ or a set; no „or“ between ranges and no expressions. Its participation limit allows a positive real number, while 11.3 says multiplicity values are integers. The base declaration of logical operations writes „ >=“ with a space inside the literal. A.3.2 also misses commas between elements in some declarations, and NOTE 3 and NOTE 4 give the two quote symbols a meaning that ISO/IEC 14977 does not have.
N-039	12.2, Figure 38 versus Tables 17-23 and Annex A	vysoká	The XOR and OR syntax of 12.2 joins the last alternative with „and“ („exactly one of Thing1, Thing2, ..., and Thingn“), as Figure 38 does („Exactly one of Safe Owner A and Safe Owner B handles Safe Opening.“) and as OPCloud does; Tables 17-23 and the Or / Xor lists of Annex A join it with „or“ („P consumes exactly one of A, B, or C.“). Both are in the norm.
N-040	12.3 versus Figure 38, Annex A.4.5.3	střední	12.3 requires agent and instrument link fans to be divergent (one enabler, several processes), but Figure 38 shows a converging agent fan (two agents, one process), Annex A defines agent and instrument Or / Xor sentences with several objects, and OPCloud draws them („Paying requires at least one of Card and Cash“).
N-041	12.5 versus 8.2.3	střední	12.5 requires event and condition versions of every XOR fan of consumption, result, effect and enabling links, and 14.2.4.1.3 says every non-control link kind has an event and a condition link; 8.2.3 states that there is no result event link and no result condition link.
N-042	12.6, Table 23	střední	The event consumption fans read „which consumes the initiated process“ and the event effect fan of Table 22 „which affects the occurring process“ (the process consumes or affects B, not itself); the stateless case of the instrument event fan still has the state („which requires s2 B“). The condition instrument fans use „requires that B is s2“ / „requires that B exists“, a form that 9.5.3.2.2 and 9.5.3.4.2 do not have.

	místo v ISO	váha	popis
N-043	12.1-12.2, Figures 36-38	nízká	Figure 37 b) declares „Central Bank handles Interest Rate Changing.“ but the process of the same OPL is „Interest Rate Raising“ (the diagram has „Interst Rate Raising“; a) has „Exchnage Rate“). The example of Figure 38 places XOR on the right and OR on the left, the captions the other way round. „Chef handles Meal Preparing“ has no full stop (Figure 36).
N-056	11.2 NOTE 2 versus Annex A.4.6.3.1, A.4.6.2.2	nízká	NOTE 2 of 11.2 says multiplicity of processes does not use participation constraints, but Annex A allows a participation constraint before process parts (part process) and tine processes.
N-057	12.4, 12.7	nízká	12.4 gives every link fan of 12.3 a state-specified version, including invocation fans, whose ends are processes without states. 12.7 states the 1/n probability for any result link to a stateful object without a state and later probability 1 for an object with one initial state, without saying which applies. The sentence prescribing the OPL of a probabilistic fan lacks a verb for its second part.
N-061	14.2.2.2 Figure 47, 9.2.2, A.4.5.3.2	nízká	The OPL of Figure 47 writes one agent sentence for two processes („Payer handles Writing & Signing and Delivering & Accepting.“, and the same for Payee). The syntax statement of 9.2.2 and the agent sentence of Annex A (A.4.5.3.2) have one process per agent sentence; no clause gives a list of processes after „handles“. The validator reads the form of the figure and reports it as I in the ISO profile (DIALECT).
N-062	12.1, Figures 11, 33, 47	střední	12.1 requires links with AND semantics (links of one kind at one process) to be written as one OPL sentence with „and“ rather than one sentence per link. The OPL of the norm's own figures does not follow it: Figure 11 has two sentences „Machining consumes ...“, Figure 33 two agent sentences of Blade Replacing (with different multiplicities, which one sentence could hardly carry), Figure 47 the agents Payer and Payee of Delivering & Accepting in two sentences. The ISO profile therefore reports separate sentences as I (OPM-12-002), the OPCloud profile as E.